Portfolio Project: Customer Churn Predictor

PSEB Artificial Intelligence Bootcamp, Business ML Activity

The business problem

A telecom company is losing customers every month, and keeping a customer costs far less than finding a new one. Your job is to find out **why customers leave**, build a model that **predicts who will leave next**, and give the company a **live tool** its team can use.

Build this as if it's going into your portfolio, because it is.

Dataset	Telco Customer Churn on Kaggle (7,043 customers, 21 columns)
You'll produce	A Colab notebook, a live Streamlit app, a GitHub repo and a LinkedIn post
AI tools	Allowed for help and debugging, but you must understand and be able to explain every line

What you'll build

Step	What you do	Why it matters
1. Load	Download the CSV from Kaggle and load it in Colab	Every project starts with the raw data
2. First look	Check the shape, data types and duplicates	Find problems before they break your model
3. Clean	Fix `TotalCharges` (stored as text, with 11 blanks) and drop `customerID`	Models need clean, numeric data
4. EDA	Plot churn by contract, tenure, monthly bill and payment method	Finds the **why** behind churn, which is the part managers care about
5. Prepare	Split the data 80/20 with `stratify=y`	Only 26.5% of customers churn, so the split must keep that ratio
6. Pipeline	Scale the numbers and one-hot encode the categories	The same steps apply to training data, test data and app inputs
7. Compare	Train Logistic Regression, Decision Tree and Random Forest	Shows which algorithm suits this problem
8. Recall	Use `class_weight="balanced"` and compare recall	A missed churner is a lost customer, so catching them matters most
9. Tune	Use GridSearchCV on the best model	Squeezes out extra performance
10. Evaluate	Confusion matrix and classification report	Proves the model works on unseen data
11. Recommend	Write 3 or 4 business actions based on your findings	Turns the analysis into decisions

Use the reference notebook `Churn_Project.ipynb` if you get stuck, but write your own explanations under each step.

Build the app

Start from `app.py`. It asks for a customer's details, shows their **churn risk** (High, Medium or Low) and suggests **actions to keep them**.

1. Download the Kaggle CSV and rename it `telco_churn.csv`.
2. Run the app and check it works. Then make it yours: change the title and wording, add an input, or add a chart.

Put it on GitHub and deploy

1. Create a **public** repo, for example `customer-churn-predictor`.
2. Upload `app.py`, `requirements.txt`, `telco_churn.csv`, your notebook and a `README.md`.
3. Go to Streamlit, then **Create app**, choose your repo and `app.py`, and click **Deploy**.
4. Add the live link to the top of your README.

README template

```
# Customer Churn Predictor
Live app: https://your-app.streamlit.app

Predicts which telecom customers are likely to leave, and suggests
actions to keep them.

## Key findings
- Month-to-month customers churn at 43%, compared with 3% on two-year contracts
- Most churn happens in the first year
- Electronic check users churn at 45%

## Model
Random Forest (balanced), catching 79% of customers who leave.

## Built with
Python, pandas, scikit-learn, Streamlit. Data from Kaggle.
```

Submit on the LMS

1. Live app link
2. GitHub repo link
3. LinkedIn post link